

Teamwork Agreement Team 7

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: Team 7

Instructor: Dr. Esperanza Linares Guerrero & Dr. Harish Chinatakunta

Program/Class: Engineering Pathway Program

Members and Roles

1. Sean Barnes - Project Manager
2. ~~Andrew Chao - Documentation/Visualization Lead~~
3. Adib Adnan - Analysis/Validation lead
4. ~~Yonathan Medina - Modeling Lead~~

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Engineering Pathway Program Discord
- Secondary channel: Team Discord Group Chat
- Response time: under 3 hours on weekdays and under 3 hours on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)
- Meeting cadence: Every Saturday 7PM (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom))
- Decision records: [Meeting notes document](#) (e.g. PM logs major decisions in a running document)

4) Collaboration & Tools

- Version control: Google drive for draft, GitHub for final (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in README.md (e.g. README.md)

File naming & style: descriptive names (e.g. Snake_case, descriptive names)

- Backups: `_firstInitialLastInitial#Revision` (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

Expected results (from GitHub)

1. At least two drone design sketches showing key geometric features
2. Results of thrust-to-weight analysis presented either in a summary table or plot
3. Results of finite element analysis
 - a. Numerical values presented in a summary table
 - b. Visualizations of x-, y-, and z-displacement and Von Mises stress
4. A final design recommendation justified using the results of your analyses

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: Brainstorm/design sketches, study quadcopter drones, Review engineering and physics concepts, set up Github repo
- Week 2: Begin programming, do CAD design
- Week 3: Run thrust-to-weight and finite element analyses, finish programming
- Week 4: Finalize report (Report should be gradually worked on), finish any unfinished tasks

(e.g.

Week 1: Problem framing, literature/background review, dataset/parameters,

Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.

Week 3: Validation and quality assurance

Week 4: Report, slides, presentation rehearsal, repository cleanup, cost analysis

7) Decision-Making & Conflict Resolution

- Default: consensus after discussion (e.g. consensus after discussion; PM facilitates

- If blocked >48 hrs: Write description of each position and hold a majority vote(e.g. Time-limited debate, majority vote, PM documents decision)
- Technical disagreements: Evidence-based comparison before vote (if needed) (e.g. Evidence-based comparison before vote)
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: Jira (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)
- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.
- Attendance: If you'll miss a meeting, give 24-hour notice and share updates asynchronously, and read meeting notes document

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- ☐ Completed MATLAB file for each simulation type (Thrust to weight & finite element)
- ☐ Drone sketches
- ☐ Drone CAD (native & common file format like STEP or STL)
- ☐ ReadMe.txt
- ☐ Hand calculation for reference (Thrust to weight)
- ☐ Cost Analysis (tentative)
- ☐ Repository organization

☐ Project report (Includes final design proposal)

(e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: Sean Barnes Date: 7/8/26

Name & Signature: Adib Adnan Date: 07/08/2026

~~Name & Signature: Andrew Chao Date: 7/8/2026~~ **Withdrew from program**

~~Name & Signature: *Jonathan Medina* Date: 7/8/2026~~ **Stopped Attending Meetings**